

Roll No.

TRA-714

**B. TECH. (SEVENTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2022
SENSOR APPLICATIONS IN MANUFACTURING**

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each question carries 10 marks.

1. (a) Discuss the basic concept terminology and functions of Sensors and Transducers. (CO1)

OR

- (b) Define the transducer dynamic and static characteristics. (CO1)

2. (a) How are the sensors and transducers classified ? (CO1)

OR

- (b) What are the major uses of operational amplifiers, filters and signal converters ? (CO1)

3. (a) How is signal processing done and in which ways it affects the performance of a sensor ? (CO1, CO2)

OR

- (b) What are the major points to be considered in using the sensors and transducers ? (CO1, CO2)

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(2)

4. (a) What are the major field applications of inductive capacitive and magnetic sensors ? (CO2)

OR

- (b) Explain the difference between sensitivity detection methods of sensors and transducers. (CO2)

5. (a) Discuss photosensors and their field of applications. (CO2)

OR

- (b) Describe the principle and workflow of Ultrasonic and Microwave sensors with a diagram. (CO2)